

Notes: Fama & French (JF, 1992)

The Cross-Section of Expected Stock Returns

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1 Big picture

- **Question.** Which firm characteristics explain the cross-section of average stock returns? In particular, does market beta explain expected returns once size, book-to-market equity, leverage, and earnings-price variables are considered?
- **Main result.** For 1963–1990 NYSE, AMEX, and NASDAQ nonfinancial stocks, the traditional market beta has little explanatory power for average returns after controlling for size and book-to-market equity.
- **Size effect.** Small firms earn higher average returns than large firms. The size premium remains visible in portfolio sorts and in Fama–MacBeth regressions.
- **Book-to-market effect.** High book-to-market equity firms earn much higher average returns than low book-to-market firms. Book-to-market is the strongest variable in the paper.
- **Beta result.** Sorting on pre-ranking beta creates large differences in post-ranking beta, but average returns are flat or even weakly declining across beta portfolios once size is controlled.
- **Leverage and earnings-price.** Leverage and earnings-price ratios are related to average returns, but their information is largely absorbed by size and book-to-market equity.
- **Economic interpretation.** Book-to-market may proxy for a distress or relative-prospects risk factor: firms with low prices relative to book values have poor prospects and high average returns.
- **Asset-pricing implication.** The single-factor CAPM does not describe the cross-section of expected returns in this sample. The evidence motivates multi-factor models built around market, size, and value factors.
- **Methodological contribution.** The paper combines portfolio sorts with Fama–MacBeth monthly cross-sectional regressions to separate the roles of beta and firm characteristics.
- **Takeaway.** Average stock returns are better summarized by size and book-to-market equity than by market beta.

2 Data and measurement

2.1 Sample construction

- The sample is the intersection of CRSP returns and the merged COMPUSTAT annual industrial files.
- The paper uses NYSE, AMEX, and NASDAQ stocks, with NASDAQ returns entering after 1972.
- Financial firms are excluded because leverage has a different economic meaning for financial firms.
- COMPUSTAT accounting data begin in 1962. Return tests run from July 1963 through December 1990.
- Accounting variables for fiscal year $t - 1$ are matched to stock returns from July of year t through June of year $t + 1$.

Interpretation: The July-to-June timing is designed to ensure that accounting data are publicly available before they are used to explain returns.

2.2 Market equity and size

Market equity is

$$ME_{it} = P_{it} \times shares_{it}.$$

Firm size in the regressions is

$$\ln(ME_{it}),$$

where ME is measured in June of year t .

- NYSE stocks define the size breakpoints.
- NYSE, AMEX, and NASDAQ firms are then assigned to size portfolios using these NYSE breakpoints.
- This prevents small NASDAQ firms from dominating the breakpoints after NASDAQ enters the sample.

Interpretation: Size is not only a control variable. It is one of the paper’s main empirical anomalies relative to the CAPM.

2.3 Book-to-market equity

Book-to-market equity is

$$\frac{BE}{ME},$$

where BE is book common equity plus balance-sheet deferred taxes, and ME is market equity.

The regression variable is

$$\ln(BE/ME).$$

Interpretation: High BE/ME means the market value of equity is low relative to accounting book equity. The paper interprets this as related to poor prospects or distress-like risk.

2.4 Leverage variables

The paper studies two leverage variables:

$$\ln(A/ME) \quad \text{and} \quad \ln(A/BE),$$

where A is total book assets.

The key accounting identity is

$$\ln(BE/ME) = \ln(A/ME) - \ln(A/BE).$$

Interpretation: Market leverage A/ME and book leverage A/BE are related to returns, but because their difference is book-to-market equity, the paper asks whether leverage adds information beyond BE/ME.

2.5 Earnings-price ratio

The paper uses:

- $E(+)/P$: earnings-price ratio for firms with positive earnings;
- an E/P dummy equal to one when earnings are negative.

Interpretation: The dummy separates negative earnings from low positive earnings. This is important because the earnings-price ratio is not meaningful in the same way when earnings are negative.

2.6 Beta construction

For the portfolio tests, the paper forms portfolios on size and pre-ranking beta. Post-ranking betas are then estimated from portfolio returns:

$$R_{pt} - R_{ft} = a_p + \beta_p(R_{Mt} - R_{ft}) + \epsilon_{pt}.$$

- Pre-ranking betas are estimated from 2 to 5 years of monthly returns before portfolio formation.
- Post-ranking betas use the full July 1963–December 1990 sample of post-ranking portfolio returns.
- This reduces measurement error in individual-stock betas.

Interpretation: The beta tests are deliberately portfolio-based because individual-stock beta estimates are noisy.

3 Models and empirical specifications

3.1 CAPM benchmark

The single-factor CAPM implies that expected excess returns are linear in market beta:

$$E(R_i) - R_f = \beta_i [E(R_M) - R_f].$$

Interpretation: If the CAPM holds, average returns should increase with beta, and variables such as size and book-to-market should not add explanatory power once beta is controlled for.

3.2 Portfolio sorts

The paper uses two-pass portfolio sorts. A typical sort is:

1. sort stocks into size deciles using NYSE breakpoints;
2. within each size decile, sort stocks into beta portfolios;
3. compute equal-weighted monthly returns from July of year t through June of year $t + 1$.

Interpretation: These sorts make the beta-return relation visible within size groups and reduce the chance that the beta result is just a disguised size effect.

3.3 Fama–MacBeth regressions

Each month, the paper runs cross-sectional regressions of the form

$$R_{it} = \gamma_{0t} + \gamma_{1t}\beta_{it} + \gamma_{2t} \ln(\text{ME}_{it}) + \gamma_{3t} \ln(\text{BE}/\text{ME})_{it} + u_{it}.$$

More general regressions add leverage and earnings-price variables:

$$R_{it} = \gamma_{0t} + \gamma_{1t}\beta_{it} + \gamma_{2t} \ln(\text{ME}_{it}) + \gamma_{3t} \ln(\text{BE}/\text{ME})_{it} \\ + \gamma_{4t} \ln(A/\text{ME})_{it} + \gamma_{5t} \ln(A/\text{BE})_{it} + \gamma_{6t} D_{E/P,it} + \gamma_{7t} E(+)/P_{it} + u_{it}.$$

The reported slope is the time-series mean:

$$\bar{\gamma}_j = \frac{1}{T} \sum_{t=1}^T \gamma_{jt}.$$

The t -statistic is

$$t(\bar{\gamma}_j) = \frac{\bar{\gamma}_j}{sd(\gamma_{jt})/\sqrt{T}}.$$

Interpretation: The Fama–MacBeth design evaluates which variables earn persistent cross-sectional premia over time.

3.4 Key empirical predictions

- **CAPM prediction:** beta should have a positive average slope and should absorb the cross-sectional variation in expected returns.
- **Size anomaly:** $\ln(\text{ME})$ should have a negative slope: smaller firms should have higher average returns.

- **Value effect:** $\ln(\text{BE}/\text{ME})$ should have a positive slope: high book-to-market firms should have higher average returns.
- **Redundancy tests:** leverage and earnings-price should lose explanatory power if their information is captured by size and book-to-market.

Interpretation: The paper’s strongest evidence is not any single sort, but the consistency between portfolio returns and Fama–MacBeth slopes.

4 Identification and empirical design

4.1 What identifies the relations

- The paper is an asset-pricing cross-sectional design, not a causal design.
- Identification comes from persistent cross-sectional covariance between firm characteristics and realized average returns.
- The monthly Fama–MacBeth procedure asks whether characteristics earn a stable average premium across 330 months.
- Portfolio sorts give a transparent nonparametric view of the same relations.

Interpretation: The estimates should be interpreted as expected-return patterns, not causal effects of size or book-to-market on returns.

4.2 Why portfolio sorts matter

- Individual-stock betas are noisy.
- Sorting on size and pre-ranking beta produces portfolios with wide spreads in post-ranking beta.
- If beta is priced, returns should vary strongly across beta portfolios within size deciles.
- The paper finds they do not.

Interpretation: This is the most direct challenge to the single-factor CAPM in the paper.

4.3 Why NYSE breakpoints matter

- NYSE breakpoints prevent the many small NASDAQ firms from dominating size and beta breakpoints after 1972.
- Applying the NYSE breakpoints to all exchanges preserves cross-exchange coverage while keeping portfolio definitions stable.

Interpretation: This design choice avoids making the evidence purely a microcap or NASDAQ composition result.

4.4 Why book-to-market is interpreted carefully

- The paper does not claim that book-to-market mechanically causes high returns.
- The authors interpret BE/ME as a summary variable related to relative distress or poor prospects.
- If book-to-market proxies for an omitted risk factor, its return premium may be rational compensation for risk.

- If not, the premium may be a mispricing or behavioral regularity.

Interpretation: The paper’s empirical contribution is stronger than its structural interpretation: it shows the relation; it does not fully explain the economic source of the value premium.

4.5 Main limitations

- The paper studies realized average returns over 1963–1990, so inference depends on this historical sample.
- COMPUSTAT coverage before 1962 is limited, which is why the main sample starts in 1963.
- The design tests characteristics and returns, but does not identify a structural risk factor directly.
- Size, book-to-market, leverage, and earnings are correlated, so interpretation of individual variables requires care.
- Later work is needed to decide whether the size and value premia are risk compensation, mispricing, or both.

Interpretation: The paper is foundational because it documents the cross-sectional facts cleanly; it leaves the theory of those facts open.

5 Tables and figures

The paper has no graph-style figures in the main text. This section directly extracts the main original tables and groups adjacent table panels to save space and improve reading speed.

5.1 Table I: Size–beta portfolios

Read:

- Panel A shows average monthly returns for 100 portfolios formed first on size and then on pre-ranking beta.
- Small firms have higher average returns than large firms: the Small-ME row has returns around 1.5–1.8%, while the Large-ME row has returns around 0.6–1.1%.
- Within size rows, returns do not increase reliably with beta.
- Panel B shows that the procedure creates large spreads in post-ranking beta.
- Panel C shows that size varies strongly across size rows but not much across beta columns within a size row.

Economic message: The key lesson is that beta variation created within size groups does not produce a strong beta premium. Size is visible; beta is not.

Table I
Average Returns, Post-Ranking β s and Average Size For Portfolios Formed on
Size and then β : Stocks Sorted on ME (Down) then Pre-Ranking β (Across):
July 1963 to December 1990

Portfolios are formed yearly. The breakpoints for the size (ME, price times shares outstanding) deciles are determined in June of year t ($t = 1963-1990$) using all NYSE stocks on CRSP. All NYSE, AMEX, and NASDAQ stocks that meet the CRSP-COMPUSTAT data requirements are allocated to the 10 size portfolios using the NYSE breakpoints. Each size decile is subdivided into 10 β portfolios using pre-ranking β s of individual stocks, estimated with 2 to 5 years of monthly returns (as available) ending in June of year t . We use only NYSE stocks that meet the CRSP-COMPUSTAT data requirements to establish the β breakpoints. The equal-weighted monthly returns on the resulting 100 portfolios are then calculated for July of year t to June of year $t + 1$.

The post-ranking β s use the full (July 1963 to December 1990) sample of post-ranking returns for each portfolio. The pre- and post-ranking β s (here and in all other tables) are the sum of the slopes from a regression of monthly returns on the current and prior month's returns on the value-weighted portfolio of NYSE, AMEX, and (after 1972) NASDAQ stocks. The average return is the time-series average of the monthly equal-weighted portfolio returns, in percent. The average size of a portfolio is the time-series average of monthly averages of $\ln(\text{ME})$ for stocks in the portfolio at the end of June of each year, with ME denominated in millions of dollars.

The average number of stocks per month for the size- β portfolios in the smallest size decile varies from 70 to 177. The average number of stocks for the size- β portfolios in size deciles 2 and 3 is between 15 and 41, and the average number for the largest 7 size deciles is between 11 and 22.

The All column shows statistics for equal-weighted size-decile (ME) portfolios. The All row shows statistics for equal-weighted portfolios of the stocks in each β group.

	All	Low- β	β -2	β -3	β -4	β -5	β -6	β -7	β -8	β -9	High- β
Panel A: Average Monthly Returns (in Percent)											
All	1.25	1.34	1.29	1.36	1.31	1.33	1.28	1.24	1.21	1.25	1.14
Small-ME	1.52	1.71	1.57	1.79	1.61	1.50	1.50	1.37	1.63	1.50	1.42
ME-2	1.29	1.25	1.42	1.36	1.39	1.65	1.61	1.37	1.31	1.34	1.11
ME-3	1.24	1.12	1.31	1.17	1.70	1.29	1.10	1.31	1.36	1.26	0.76
ME-4	1.25	1.27	1.13	1.54	1.06	1.34	1.06	1.41	1.17	1.35	0.98
ME-5	1.29	1.34	1.42	1.39	1.48	1.42	1.18	1.13	1.27	1.18	1.08
ME-6	1.17	1.08	1.53	1.27	1.15	1.20	1.21	1.18	1.04	1.07	1.02
ME-7	1.07	0.95	1.21	1.26	1.09	1.18	1.11	1.24	0.62	1.32	0.76
ME-8	1.10	1.09	1.05	1.37	1.20	1.27	0.98	1.18	1.02	1.01	0.94
ME-9	0.95	0.98	0.88	1.02	1.14	1.07	1.23	0.94	0.82	0.88	0.59
Large-ME	0.89	1.01	0.93	1.10	0.94	0.93	0.89	1.03	0.71	0.74	0.56

Table I, Panel A: average returns for size–beta portfolios.

Table I—Continued

	All	Low- β	β -2	β -3	β -4	β -5	β -6	β -7	β -8	β -9	High- β
Panel B: Post-Ranking β s											
All		0.87	0.99	1.09	1.16	1.26	1.29	1.35	1.45	1.52	1.72
Small-ME	1.44	1.05	1.18	1.28	1.32	1.40	1.40	1.49	1.61	1.64	1.79
ME-2	1.39	0.91	1.15	1.17	1.24	1.36	1.41	1.43	1.50	1.66	1.76
ME-3	1.35	0.97	1.13	1.13	1.21	1.26	1.28	1.39	1.50	1.51	1.75
ME-4	1.34	0.78	1.03	1.17	1.16	1.29	1.37	1.46	1.51	1.64	1.71
ME-5	1.25	0.66	0.85	1.12	1.15	1.16	1.26	1.30	1.43	1.59	1.68
ME-6	1.23	0.61	0.78	1.05	1.16	1.22	1.28	1.36	1.46	1.49	1.70
ME-7	1.17	0.57	0.92	1.01	1.11	1.14	1.26	1.24	1.39	1.34	1.60
ME-8	1.09	0.53	0.74	0.94	1.02	1.13	1.12	1.18	1.26	1.35	1.52
ME-9	1.03	0.58	0.74	0.80	0.95	1.06	1.15	1.14	1.21	1.22	1.42
Large-ME	0.92	0.57	0.71	0.78	0.89	0.95	0.92	1.02	1.01	1.11	1.32
Panel C: Average Size (ln(ME))											
All	4.11	3.86	4.26	4.33	4.41	4.27	4.32	4.26	4.19	4.03	3.77
Small-ME	2.24	2.12	2.27	2.30	2.30	2.28	2.29	2.30	2.32	2.25	2.15
ME-2	3.63	3.65	3.68	3.70	3.72	3.69	3.70	3.69	3.69	3.70	3.68
ME-3	4.10	4.14	4.18	4.12	4.15	4.16	4.16	4.18	4.14	4.15	4.15
ME-4	4.50	4.53	4.53	4.57	4.54	4.56	4.55	4.52	4.58	4.52	4.56
ME-5	4.89	4.91	4.91	4.93	4.95	4.93	4.92	4.93	4.92	4.92	4.95
ME-6	5.30	5.30	5.33	5.34	5.34	5.33	5.33	5.33	5.33	5.34	5.36
ME-7	5.73	5.73	5.75	5.77	5.76	5.73	5.77	5.77	5.76	5.72	5.76
ME-8	6.24	6.26	6.27	6.26	6.24	6.24	6.27	6.24	6.24	6.24	6.26
ME-9	6.82	6.82	6.84	6.82	6.82	6.81	6.81	6.81	6.81	6.80	6.83
Large-ME	7.93	7.94	8.04	8.10	8.04	8.02	8.02	7.94	7.80	7.75	7.62

Table I, Panels B–C: post-ranking betas and average size.

5.2 Table II: Portfolios formed on size or beta alone

Read:

- Panel A forms portfolios on size. Average returns decline from 1.64% for the smallest group to 0.90% for the largest group.
- Panel A also shows that small firms have high book-to-market ratios and higher leverage-related characteristics.
- Panel B forms portfolios on pre-ranking beta. Post-ranking beta rises from about 0.81 to 1.73, but average returns remain flat, around 1.18–1.33%.
- The beta portfolios have no clear increasing pattern in average returns.

Economic message: Sorting on beta alone produces large beta spreads, but not a beta-return relation. Sorting on size produces a clear return relation.

Table II
Properties of Portfolios Formed on Size or Pre-Ranking β :
July 1963 to December 1990

At the end of June of each year t , 12 portfolios are formed on the basis of ranked values of size (ME) or pre-ranking β . The pre-ranking β s use 2 to 5 years (as available) of monthly returns ending in June of t . Portfolios 2–9 cover deciles of the ranking variables. The bottom and top 2 portfolios (1A, 1B, 10A, and 10B) split the bottom and top deciles in half. The breakpoints for the ME portfolios are based on ranked values of ME for all NYSE stocks on CRSP. NYSE breakpoints for pre-ranking β s are also used to form the β portfolios. NYSE, AMEX, and NASDAQ stocks are then allocated to the size or β portfolios using the NYSE breakpoints. We calculate each portfolio's monthly equal-weighted return for July of year t to June of year $t + 1$, and then reform the portfolios in June of $t + 1$.

BE is the book value of common equity plus balance-sheet deferred taxes, A is total book assets, and E is earnings (income before extraordinary items, plus income-statement deferred taxes, minus preferred dividends). BE, A, and E are for each firm's latest fiscal year ending in calendar year $t - 1$. The accounting ratios are measured using market equity ME in December of year $t - 1$. Firm size $\ln(\text{ME})$ is measured in June of year t , with ME denominated in millions of dollars.

The average return is the time-series average of the monthly equal-weighted portfolio returns, in percent. $\ln(\text{ME})$, $\ln(\text{BE}/\text{ME})$, $\ln(\text{A}/\text{ME})$, $\ln(\text{A}/\text{BE})$, E/P, and E/P dummy are the time-series averages of the monthly average values of these variables in each portfolio. Since the E/P dummy is 0 when earnings are positive, and 1 when earnings are negative, E/P dummy gives the average proportion of stocks with negative earnings in each portfolio.

β is the time-series average of the monthly portfolio β s. Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of June of year t (Table I). These individual-firm β s are averaged to compute the monthly β s for each portfolio for July of year t to June of year $t + 1$.

Firms is the average number of stocks in the portfolio each month.

	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel A: Portfolios Formed on Size												
Return	1.64	1.16	1.29	1.24	1.25	1.29	1.17	1.07	1.10	0.95	0.88	0.90
β	1.44	1.44	1.39	1.34	1.33	1.24	1.22	1.16	1.08	1.02	0.95	0.90
$\ln(\text{ME})$	1.98	3.18	3.63	4.10	4.50	4.89	5.30	5.73	6.24	6.82	7.39	8.44
$\ln(\text{BE}/\text{ME})$	-0.01	-0.21	-0.23	-0.26	-0.32	-0.36	-0.36	-0.44	-0.40	-0.42	-0.51	-0.65
$\ln(\text{A}/\text{ME})$	0.73	0.50	0.46	0.43	0.37	0.32	0.32	0.24	0.29	0.27	0.17	-0.03
$\ln(\text{A}/\text{BE})$	0.75	0.71	0.69	0.69	0.68	0.67	0.68	0.67	0.69	0.70	0.68	0.62
E/P dummy	0.26	0.14	0.11	0.09	0.06	0.04	0.04	0.03	0.03	0.02	0.02	0.01
E(+)/P	0.09	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.09	0.09
Firms	772	189	236	170	144	140	128	125	119	114	60	64

Table II, Panel A: portfolios formed on size.

Table II—Continued												
	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel B: Portfolios Formed on Pre-Ranking β												
Return	1.20	1.20	1.32	1.26	1.31	1.30	1.30	1.23	1.23	1.33	1.34	1.18
β	0.81	0.79	0.92	1.04	1.13	1.19	1.26	1.32	1.41	1.52	1.63	1.73
$\ln(\text{ME})$	4.21	4.86	4.75	4.68	4.59	4.48	4.36	4.25	3.97	3.78	3.52	3.15
$\ln(\text{BE}/\text{ME})$	-0.18	-0.13	-0.22	-0.21	-0.23	-0.22	-0.22	-0.25	-0.23	-0.27	-0.31	-0.50
$\ln(\text{A}/\text{ME})$	0.60	0.66	0.49	0.45	0.42	0.42	0.45	0.42	0.47	0.46	0.46	0.31
$\ln(\text{A}/\text{BE})$	0.78	0.79	0.71	0.66	0.64	0.65	0.67	0.67	0.70	0.73	0.77	0.81
E/P dummy	0.12	0.06	0.09	0.09	0.08	0.09	0.10	0.12	0.12	0.14	0.17	0.23
E(+)/P	0.11	0.12	0.10	0.10	0.10	0.10	0.10	0.09	0.10	0.09	0.09	0.08
Firms	116	80	185	181	179	182	185	205	227	267	165	291

Table II, Panel B: portfolios formed on pre-ranking beta.

5.3 Table III: Fama–MacBeth regressions

Read:

- The average slope on beta alone is 0.15 with a t -statistic of 0.46, so beta is not reliably priced.
- The average slope on $\ln(\text{ME})$ alone is -0.15 with $t = -2.58$.
- The average slope on $\ln(\text{BE}/\text{ME})$ alone is 0.50 with $t = 5.71$.
- In the bivariate size and book-to-market regression, size remains negative and book-to-market remains positive.
- In the full regressions, beta remains weak, while book-to-market remains economically and statistically important.

Economic message: Table III is the core regression evidence: size and book-to-market explain the cross-section; beta does not.

Table III
Average Slopes (*t*-Statistics) from Month-by-Month Regressions of
Stock Returns on β , Size, Book-to-Market Equity, Leverage, and E/P:
July 1963 to December 1990

Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of June of year t (Table I). BE is the book value of common equity plus balance-sheet deferred taxes, A is total book assets, and E is earnings (income before extraordinary items, plus income-statement deferred taxes, minus preferred dividends). BE, A, and E are for each firm's latest fiscal year ending in calendar year $t - 1$. The accounting ratios are measured using market equity ME in December of year $t - 1$. Firm size $\ln(\text{ME})$ is measured in June of year t . In the regressions, these values of the explanatory variables for individual stocks are matched with CRSP returns for the months from July of year t to June of year $t + 1$. The gap between the accounting data and the returns ensures that the accounting data are available prior to the returns. If earnings are positive, $E(+)/P$ is the ratio of total earnings to market equity and E/P dummy is 0. If earnings are negative, $E(+)/P$ is 0 and E/P dummy is 1.

The average slope is the time-series average of the monthly regression slopes for July 1963 to December 1990, and the *t*-statistic is the average slope divided by its time-series standard error.

On average, there are 2267 stocks in the monthly regressions. To avoid giving extreme observations heavy weight in the regressions, the smallest and largest 0.5% of the observations on $E(+)/P$, BE/ME, A/ME, and A/BE are set equal to the next largest or smallest values of the ratios (the 0.005 and 0.995 fractiles). This has no effect on inferences.

β	$\ln(\text{ME})$	$\ln(\text{BE}/\text{ME})$	$\ln(\text{A}/\text{ME})$	$\ln(\text{A}/\text{BE})$	E/P Dummy	$E(+)/P$
0.15 (0.46)						
	-0.15 (-2.58)					
-0.37 (-1.21)	-0.17 (-3.41)					
		0.50 (5.71)				
			0.50 (5.69)	-0.57 (-5.34)		
					0.57 (2.28)	4.72 (4.57)
	-0.11 (-1.99)	0.35 (4.44)				
	-0.11 (-2.06)		0.35 (4.32)	-0.50 (-4.56)		
	-0.16 (-3.06)				0.06 (0.38)	2.99 (3.04)
	-0.13 (-2.47)	0.33 (4.46)			-0.14 (-0.90)	0.87 (1.23)
	-0.13 (-2.47)		0.32 (4.28)	-0.46 (-4.45)	-0.08 (-0.56)	1.15 (1.57)

*Table III: average slopes and *t*-statistics from monthly Fama-MacBeth regressions.*

5.4 Table IV: Book-to-market and earnings-price portfolios

Read:

- Panel A sorts stocks by book-to-market equity. Average returns rise strongly from low BE/ME to high BE/ME portfolios.
- The low book-to-market portfolio has an average return of about 0.30%, while the high book-to-market portfolio has an average return around 1.83%.
- Panel B sorts by earnings-price. Positive earnings-price portfolios show a return pattern, but negative

earnings and size effects complicate the simple relation.

- Post-ranking betas vary little across these portfolios, so beta does not explain the value premium.

Economic message: Book-to-market captures a strong cross-sectional return pattern that is not just a beta pattern.

Table IV
Properties of Portfolios Formed on Book-to-Market Equity (BE/ME) and Earnings-Price Ratio (E/P):
July 1963 to December 1990

At the end of each year $t - 1$, 12 portfolios are formed on the basis of ranked values of BE/ME or E/P. Portfolios 2–9 cover deciles of the ranking variables. The bottom and top 2 portfolios (1A, 1B, 10A, and 10B) split the bottom and top deciles in half. For E/P, there are 13 portfolios; portfolio 0 is stocks with negative E/P. Since BE/ME and E/P are not strongly related to exchange listing, their portfolio breakpoints are determined on the basis of the ranked values of the variables for all stocks that satisfy the CRSP-COMPSTAT data requirements. BE is the book value of common equity plus balance-sheet deferred taxes, A is total book assets, and E is earnings (income before extraordinary items, plus income-statement deferred taxes, minus preferred dividends). BE, A, and E are for each firm's latest fiscal year ending in calendar year $t - 1$. The accounting ratios are measured using market equity ME in December of year $t - 1$. Firm size $\ln(\text{ME})$ is measured in June of year t , with ME denominated in millions of dollars. We calculate each portfolio's monthly equal-weighted return for July of year t to June of year $t + 1$, and then reform the portfolios at the end of year t .

Return is the time-series average of the monthly equal-weighted portfolio returns (in percent). $\ln(\text{ME})$, $\ln(\text{BE}/\text{ME})$, $\ln(\text{A}/\text{ME})$, $\ln(\text{A}/\text{BE})$, $\text{E}(+)/\text{P}$, and E/P dummy are the time-series averages of the monthly average values of these variables in each portfolio. Since the E/P dummy is 0 when earnings are positive, and 1 when earnings are negative, E/P dummy gives the average proportion of stocks with negative earnings in each portfolio.

β is the time-series average of the monthly portfolio β s. Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of June of year t (Table I). These individual-firm β s are averaged to compute the monthly β s for each portfolio for July of year t to June of year $t + 1$. Firms is the average number of stocks in the portfolio each month.

Portfolio	0	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel A: Stocks Sorted on Book-to-Market Equity (BE/ME)													
Return	0.30	0.67	0.87	0.97	1.04	1.17	1.30	1.44	1.50	1.59	1.92	1.83	
β	1.36	1.34	1.32	1.30	1.28	1.27	1.27	1.27	1.27	1.29	1.33	1.35	
$\ln(\text{ME})$	4.53	4.67	4.69	4.56	4.47	4.38	4.23	4.06	3.85	3.51	3.06	2.65	
$\ln(\text{BE}/\text{ME})$	-2.22	-1.51	-1.09	-0.75	-0.51	-0.32	-0.14	0.03	0.21	0.42	0.66	1.02	
$\ln(\text{A}/\text{ME})$	-1.24	-0.79	-0.40	-0.05	0.20	0.40	0.56	0.71	0.91	1.12	1.35	1.75	
$\ln(\text{A}/\text{BE})$	0.94	0.71	0.68	0.70	0.71	0.71	0.70	0.68	0.70	0.70	0.70	0.73	
E/P dummy	0.29	0.15	0.10	0.08	0.08	0.08	0.09	0.09	0.11	0.15	0.22	0.36	
$\text{E}(+)/\text{P}$	0.03	0.04	0.06	0.08	0.09	0.10	0.11	0.11	0.12	0.12	0.11	0.10	
Firms	89	98	209	222	226	230	235	237	239	239	120	117	

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Table IV, Panel A: stocks sorted on book-to-market equity.

Table IV—Continued													
Portfolio	0	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel B: Stocks Sorted on Earnings-Price Ratio (E/P)													
Return	1.46	1.04	0.93	0.94	1.03	1.18	1.22	1.33	1.42	1.46	1.57	1.74	1.72
β	1.47	1.40	1.35	1.31	1.28	1.26	1.25	1.26	1.24	1.23	1.24	1.28	1.31
$\ln(\text{ME})$	2.48	3.64	4.33	4.61	4.64	4.63	4.58	4.49	4.37	4.28	4.07	3.82	3.52
$\ln(\text{BE}/\text{ME})$	-0.10	-0.76	-0.91	-0.79	-0.61	-0.47	-0.33	-0.21	-0.08	0.02	0.15	0.26	0.40
$\ln(\text{A}/\text{ME})$	0.90	-0.05	-0.27	-0.16	0.03	0.18	0.31	0.44	0.58	0.70	0.85	1.01	1.25
$\ln(\text{A}/\text{BE})$	0.99	0.70	0.63	0.63	0.64	0.65	0.64	0.65	0.66	0.68	0.71	0.75	0.86
E/P dummy	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
$\text{E}(+)/\text{P}$	0.00	0.01	0.03	0.05	0.06	0.08	0.09	0.11	0.12	0.14	0.16	0.20	0.28
Firms	355	88	90	182	190	193	196	194	197	195	195	95	91

Table IV, Panel B: stocks sorted on earnings-price ratio.

5.5 Table V and Table VI: Size-value matrix and subperiod robustness

Read:

- Table V forms 100 portfolios on size and book-to-market equity.
- Within nearly every size row, high book-to-market portfolios earn higher average returns than low book-to-market portfolios.
- The All row rises from 0.64% for low BE/ME to 1.63% for high BE/ME.

The Cross-Section of Expected Stock Returns

- Table VI shows that the book-to-market slope is positive in both subperiods, 1963–1976 and 1977–1990.
- The beta slopes remain weak across subperiods.

Economic message: The value effect is not confined to one size group or one subperiod. It is the most robust characteristic relation in the paper.

Table V

Average Monthly Returns on Portfolios Formed on Size and Book-to-Market Equity; Stocks Sorted by ME (Down) and then BE/ME (Across): July 1963 to December 1990

In June of each year t , the NYSE, AMEX, and NASDAQ stocks that meet the CRSP-COMPUSTAT data requirements are allocated to 10 size portfolios using the NYSE size (ME) breakpoints. The NYSE, AMEX, and NASDAQ stocks in each size decile are then sorted into 10 BE/ME portfolios using the book-to-market ratios for year $t - 1$. BE/ME is the book value of common equity plus balance-sheet deferred taxes for fiscal year $t - 1$, over market equity for December of year $t - 1$. The equal-weighted monthly portfolio returns are then calculated for July of year t to June of year $t + 1$.

Average monthly return is the time-series average of the monthly equal-weighted portfolio returns (in percent).

The All column shows average returns for equal-weighted size decile portfolios. The All row shows average returns for equal-weighted portfolios of the stocks in each BE/ME group.

	Book-to-Market Portfolios										
	All	Low	2	3	4	5	6	7	8	9	High
All	1.23	0.64	0.98	1.06	1.17	1.24	1.26	1.39	1.40	1.50	1.63
Small-ME	1.47	0.70	1.14	1.20	1.43	1.56	1.51	1.70	1.71	1.82	1.92
ME-2	1.22	0.43	1.05	0.96	1.19	1.33	1.19	1.58	1.28	1.43	1.79
ME-3	1.22	0.56	0.88	1.23	0.95	1.36	1.30	1.30	1.40	1.54	1.60
ME-4	1.19	0.39	0.72	1.06	1.36	1.13	1.21	1.34	1.59	1.51	1.47
ME-5	1.24	0.88	0.65	1.08	1.47	1.13	1.43	1.44	1.26	1.52	1.49
ME-6	1.15	0.70	0.98	1.14	1.23	0.94	1.27	1.19	1.19	1.24	1.50
ME-7	1.07	0.95	1.00	0.99	0.83	0.99	1.13	0.99	1.16	1.10	1.47
ME-8	1.08	0.66	1.13	0.91	0.95	0.99	1.01	1.15	1.05	1.29	1.55
ME-9	0.95	0.44	0.89	0.92	1.00	1.05	0.93	0.82	1.11	1.04	1.22
Large-ME	0.89	0.93	0.88	0.84	0.71	0.79	0.83	0.81	0.96	0.97	1.18

Table V: size by book-to-market average return matrix.

Table VI

Subperiod Average Monthly Returns on the NYSE Equal-Weighted and Value-Weighted Portfolios and Subperiod Means of the Intercepts and Slopes from the Monthly FM Cross-Sectional Regressions of Returns on (a) Size ($\ln(\text{ME})$) and Book-to-Market Equity ($\ln(\text{BE}/\text{ME})$), and (b) β , $\ln(\text{ME})$, and $\ln(\text{BE}/\text{ME})$

Mean is the time-series mean of a monthly return, Std is its time-series standard deviation, and $t(\text{Mn})$ is Mean divided by its time-series standard error.

Variable	7/63-12/90 (330 Mos.)			7/63-12/76 (162 Mos.)			1/77-12/90 (168 Mos.)		
	Mean	Std	$t(\text{Mn})$	Mean	Std	$t(\text{Mn})$	Mean	Std	$t(\text{Mn})$
NYSE Value-Weighted (VW) and Equal-Weighted (EW) Portfolio Returns									
VW	0.81	4.47	3.27	0.56	4.26	1.67	1.04	4.66	2.89
EW	0.97	5.49	3.19	0.77	5.70	1.72	1.15	5.28	2.82
$R_{it} = a + b_{2t}\ln(\text{ME}_{it}) + b_{3t}\ln(\text{BE}/\text{ME}_{it}) + e_{it}$									
a	1.77	8.51	3.77	1.86	10.10	2.33	1.69	6.67	3.27
b_2	-0.11	1.02	-1.99	-0.16	1.25	-1.62	-0.07	0.73	-1.16
b_3	0.35	1.45	4.43	0.36	1.53	2.96	0.35	1.37	3.30
$R_{it} = a + b_{1t}\beta_{it} + b_{2t}\ln(\text{ME}_{it}) + b_{3t}\ln(\text{BE}/\text{ME}_{it}) + e_{it}$									
a	2.07	5.75	6.55	1.73	6.22	3.54	2.40	5.25	5.92
b_1	-0.17	5.12	-0.62	0.10	5.33	0.25	-0.44	4.91	-1.17
b_2	-0.12	0.89	-2.52	-0.15	1.03	-1.91	-0.09	0.74	-1.64
b_3	0.33	1.24	4.80	0.34	1.36	3.17	0.31	1.10	3.67

Table VI: subperiod Fama-MacBeth slopes.

6 Short summary

- Fama and French study the cross-section of average returns for NYSE, AMEX, and NASDAQ nonfinancial firms from July 1963 to December 1990.
- The paper tests beta, size, book-to-market equity, leverage, and earnings-price variables.
- Portfolio sorts show a strong size effect and a strong book-to-market effect.
- Sorting on beta creates large beta spreads but does not create a strong return spread.
- Fama-MacBeth regressions show that beta has little explanatory power.
- The slope on size is negative: small firms earn higher average returns.
- The slope on book-to-market equity is positive and highly significant.
- Leverage and earnings-price variables are related to returns, but much of their information is captured by book-to-market and size.
- The evidence rejects the simple CAPM description of expected returns.
- The paper motivates the later Fama-French factor model, especially the size and value factors.

7 Conclusion

7.1 Core conclusion

Fama and French show that the cross-section of average stock returns is not well explained by market beta. Size and book-to-market equity do much more work. The evidence is visible in both portfolio sorts and monthly Fama-MacBeth regressions.

7.2 Economic intuition

The intuition is that firm characteristics summarize economically important differences that beta misses. Small firms and high book-to-market firms have high average returns. High book-to-market firms may be firms with poor prospects, distress, or exposure to a systematic risk factor not captured by the market beta alone.

7.3 Takeaway

The paper changes empirical asset pricing by shifting attention from a single market-beta dimension to multiple cross-sectional characteristics. Its main message is simple: in the data, expected returns vary with size and value much more than with CAPM beta.